

Spiral Similarity Pset

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1 Buffer Problems

Problem 1

Let $ABCD$ be a cyclic quadrilateral with circumcenter O lying in the interior. Let E and F be the midpoints of sides BC and AD respectively. Let X be the point lying on the same side of the line EF as vertex C such that $\triangle EXF$ and $\triangle BOA$ are similar. Prove that, $XC = XD$.

Problem 2

Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that, the circumcircles of triangles SAE , SBF , TCF and TDE pass through a common point.

Problem 3

Let $ABCD$ be a convex quadrilateral. X lies in the interior of $ABCD$ such that $\triangle AXB \sim \triangle DXC$. Y is another point in the interior such that $AY = DY$ and $BY = CY$. Prove that, $\angle AYD = 2\angle ABX$.

Problem 4

Quadrilateral $ABCD$ is given with $AD \nparallel BC$. The midpoints of AD and BC are denoted by M and N respectively. The line MN intersects the lines AC and BD in points K and L respectively. Prove that, the circumcircles of $\triangle AKM$ and $\triangle BNL$ have a common point on AB .

Problem 5 (If you have solved this, feel proud)

Let the excircle of triangle ABC opposite the vertex A be tangent to the side BC at the point A_1 . Define the points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C , respectively. Suppose that the circumcentre of triangle $A_1B_1C_1$ lies on the circumcircle of triangle ABC . Prove that triangle ABC is right-angled.